
AWS Cloud9 - Overview

What is AWS Cloud9?

AWS Cloud9 is a **cloud-based integrated development environment (IDE)** that lets you write, run, and debug code directly in your browser. It comes prepackaged with essential tools for most programming languages, removing the need to install anything locally.

Key Features

Feature	Description
Cloud-based IDE	Access from anywhere using a browser. No local setup required.
Multiple Languages	Supports Python, JavaScript, PHP, Ruby, Go, C++, and more.
Built-in Terminal	Comes with a Linux shell and AWS CLI preinstalled.
Code Collaboration	Share environments for real-time pair programming.
EC2 Integration	Uses Amazon EC2 as the compute backend.
Environment Customization	Easily customize the environment with additional tools or configurations.
Access to AWS Resources	Built-in credentials allow you to access AWS services securely.

Cloud9 Interface Components

- **File Explorer** – Navigate project files and folders.

- **Editor** – Code editor with syntax highlighting and auto-completion.
 - **Terminal** – Command-line interface for managing your environment.
 - **Run/Debug Buttons** – Quickly run or debug applications.
 - **Collaborators Panel** – Invite teammates and see what they're doing in real time.
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Typical Use Cases

- Cloud-based software development.
 - Remote team coding and collaboration.
 - Teaching and learning to code.
 - Quickly prototyping cloud-native applications.
 - Working with serverless frameworks like AWS Lambda.
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Security & Permissions

- Uses **IAM roles** and policies to manage access.
 - Supports **resource-level permissions**.
 - Can be integrated with **AWS SSO** and **IAM Identity Center** for enterprise setups.
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Benefits of AWS Cloud9

- **Zero local setup:** Get started immediately in a preconfigured environment.
 - **Portability:** Work from any internet-connected device.
 - **Collaboration:** Work with team members in real time.
 - **AWS integration:** Seamlessly deploy to Lambda, EC2, and other services.
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How to Get Started

1. Go to the [AWS Cloud9 Console](#).
 2. Click **Create environment**.
 3. Name your environment and provide a description.
 4. Choose an instance type and platform (Amazon EC2 or SSH).
 5. Click **Create** — your IDE will launch in a browser tab.
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Pricing

- **Cloud9 itself is free** — you pay only for the **underlying compute (EC2)** and **storage (EBS)**.
 - Use **t2.micro** or **t3.micro** (free tier eligible) for minimal cost.
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